

LIM LI JING A25 CS 0248
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SECI1013: DISCRETE STRUCTURES

SESSION 2025/2026 – SEMESTER 1

ASSIGNMENT 1 (CHAPTER 1 – SET THEORY & LOGIC)

INSTRUCTIONS:

- This assignment must be conducted in a group. Please clearly write the group members' names & matric numbers on the front page of the submission.
- Solutions for each question must be readable and neatly written on plain A4 paper. Every step or calculation should be properly shown. Failure to do so will result in the rejection of the submission of the assignment.
- This assignment consist of 6 questions (31 marks), contributing 5% of overall course marks.

Question 1

[5 marks]

Let U be the set Z of all integer numbers and let A, B, C are the subset of U . Given $A = \{x \text{ is a solution of } x^2 + x - 6 = 0\}$, $B = \{-3, 1, 2, 4\}$ and $C = \{x \in Z | 2 \leq x < 5\}$. Compute :

- A'
- $(B - A) \cap C$.
- $|P(B \cap C)|$

Question 2

[6 marks]

Let P, Q and R are sets, prove that $((P \cup Q) \cap R)' \cup Q' = Q \cap R$ by showing all laws that used.

Question 3

[4 marks]

There are 35 students in the art class and 57 students in the science class. Find the number of students who are either in art class or in science class.

- When two classes meet at different hours and 12 students are enrolled in both activities. (2marks)
- When two classes meet at the same hour (2 marks)

Question 4**[6 marks]**

Consider the statement:

"If you try hard and you have a talent then you will get rich"

- a) Translate the statement into logic symbols. Use p , q and r to represent the propositions. Clearly state which statement is p , q and r .
- b) Suppose you found out that the statement was a lie even you try hard. What can you conclude?
- c) If you are rich but you do not try hard or have talent. Was the statement true or false? Support your conclusion.

Question 5**[5 marks]**

Use truth table to check if the compound propositions A and B are logically equivalent.

$$A = \neg(p \vee (q \wedge (r \rightarrow p)))$$

$$B = \neg p \wedge (q \rightarrow r)$$

Question 6**[5 marks]**

Proof that if x is an odd integer and y is an even integer then $x^2 - 2y$ is an odd integer using direct proofing

$$A = \{2, -3\}$$

$$B = \{-3, 1, 2, 4\}$$

$$C = \{2, 3, 4\}$$

$$(a) A' = \{x \in \mathbb{Z} \mid x \neq -3, x \neq -2\}$$

$$(b) (B - A) \cap C$$

$$B - A = \{1, 4\}$$

$$(B - A) \cap C = \{4\}$$

$$(c) B \cap C = \{2, 4\}$$

$$|B \cap C| = 2$$

$$|P(B \cap C)| = 2^2$$

$$|P(B \cap C)| = 4$$

$$P(B \cap C) = \{\emptyset, \{2\}, \{4\}, \{2, 4\}\}$$

$$2. ((C \cup Q) \wedge R)' \cup Q')'$$

$$= (((C \cup Q) \wedge R)')' \wedge (Q')' \quad [\text{De Morgan's laws}]$$

$$= (C \cup Q) \wedge R \wedge Q \quad [\text{Double Complement laws}]$$

$$= Q \wedge (C \cup Q) \wedge R \quad [\text{Commutative laws}]$$

$$= (Q \wedge (C \cup Q)) \wedge R \quad [\text{Associative laws}]$$

$$= Q \wedge R \quad [\text{Absorption laws}]$$

$$= Q \wedge R \quad [\text{Proven}]$$

let A = students in the art class
 B = students in the science class

(a)

$$|A| = 35$$

$$|B| = 57$$

$$|A \cap B| = 12$$

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| \\ &= 35 + 57 - 12 \\ &= 80 \end{aligned}$$

(b)

$$|A| = 35$$

$$|B| = 57$$

$$|A \cap B| = 0$$

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| \\ &= 35 + 57 - 0 \\ &= 92 \end{aligned}$$

4. (a)

p = you try hard

q = you have a talent

r = you will get rich

$$\therefore (p \wedge q) \rightarrow r$$

(b)

$(p \wedge q) \rightarrow r$ is a statement called a conditional proposition. The statement will become false if $p \wedge q$ is True and r is false. ~~Since 'you try hard' which means p is true,~~ From the scenario, "you try hard" which means p is true and "the statement was a lie" which means the statement is false and r is false. Since p is true, $(p \wedge q)$ is true, then q must be true (you have a talent). Since r is false, you will not get rich.

$(p \wedge q) = \text{true}$ $r = \text{False}$
 $\therefore$ You have a talent, but you will not get rich.

c)

The statement is true. Since p (you try hard) is false and q (you have a talent) is false, $(p \wedge q)$ is false and r (you are rich) is true. A conditional statement is only false when $p \wedge q$ is true and the r is false.

p	q	r	$p \wedge q$	$(p \wedge q) \rightarrow r$
F	F	T	F	T

P	q	r	$r \rightarrow p$	$q \wedge (r \rightarrow p)$	$p \vee (q \wedge (r \rightarrow p))$	$\neg p$	$q \rightarrow r$	A	B
T	T	T	T	T	T	F	F	F	F
T	T	F	T	T	T	F	F	F	F
T	F	T	T	F	T	F	F	F	F
T	F	F	T	F	T	F	F	F	F
F	T	T	F	F	F	T	T	T	T
F	T	F	T	T	T	T	T	F	F
F	F	T	F	F	F	T	T	T	T
F	F	F	T	F	F	T	T	T	T

$\therefore$ The truth table shows that, $A \equiv B$, A and B are logically equivalent.

6. $p(x) = x$ is an odd integer
 $q(y) = y$ is an even integer

$R(x, y) = x^2 - 2y$ is an odd integer

$$(p(x) \wedge q(y)) \rightarrow R(x, y)$$

$$\text{let } x = 2n+1$$

$$\text{let } y = 2m$$

$$x^2 - 2y = (2n+1)^2 - 2(2m)$$

$$= 4n^2 + 4n + 1 - 4m$$

$$= 4n^2 + 4n - 4m + 1$$

$$= 2(2n^2 + 2n - 2m) + 1 \quad \text{let } k = 2n^2 + 2n - 2m$$

$$= 2k + 1 \quad (\text{odd})$$

$\therefore x^2 - 2y$ is odd integer, so $p(x)$ and $q(y)$ is true.
 Overall the statement is true.